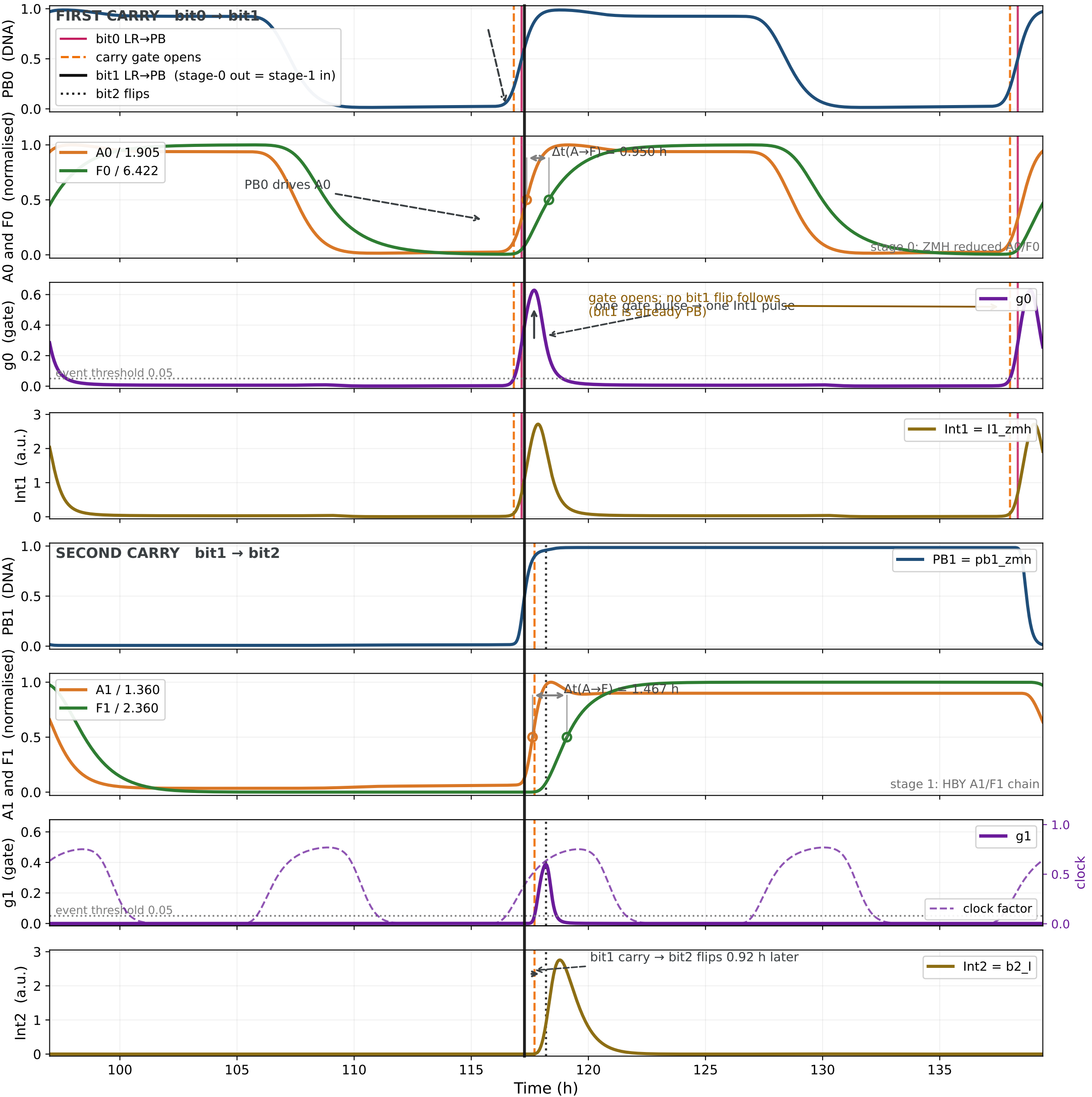


Two-stage carry timing, HBY-ZMH-HBY (HZH, 34-state)

Han v53d upstream (peak_load_fraction = 0); frozen selected_v1 + HbyConfig; 300 h, 1 min grid, DOP853 rtol 2e-7/atol 2e-9; criterion HZH_MOD8_CAUSAL_V1.
A/F curves are each normalised to their own maximum inside the window; the number after the slash in each legend entry is that maximum.



window 97.0–139.4 h = one complete second-stage carry period | low-order carries 2, high-order carry 1
 $\Delta t(A \rightarrow F)$ stage 0 0.950 h (ZMH reduced A0/F0) stage 1 1.467 h (HBY A1/F1, +54%)
gate peak g0 0.629 g1 0.387 | g0 opens -0.28 h before the J_rev0 peak; g1 opens 0.50 h after the J_rev1 peak